

Chapter 5. Object Recognition from Visual Appearance

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Chapter 5. Object Recognition from Visual Appearance

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Chapter 5. Object Recognition from Visual Appearance

5.1 Overview

We will study object recognition of 3D objects from their visual appearance in 2D images.

This is referred as appearance matching.

There are two basic approaches for objects representation:

- Shape: the explicit representation of the 3D geometry of the object.
- Appearance: depends on the pose, illumination.

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5.1 Overview

An image set is then obtained represented the appearance of the object. The reduction of the dimensionality is necessary in order to achieve the matching.

Using PCA, we lower dimensional subspace in which we can represent the original image set by simply projecting the image set.

We end up for each object , a compact parametric representation of the object.

We terminate with a pipeline for appearance matching

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5.2 Shape vs. Appearance

Basic approaches to represent 3D object.

The explicit representation of the three dimensional geometry of an object is used in different domains: manufacturing, factory automation, etc.

Example of such representation: Voxel, useful in computer graphics.

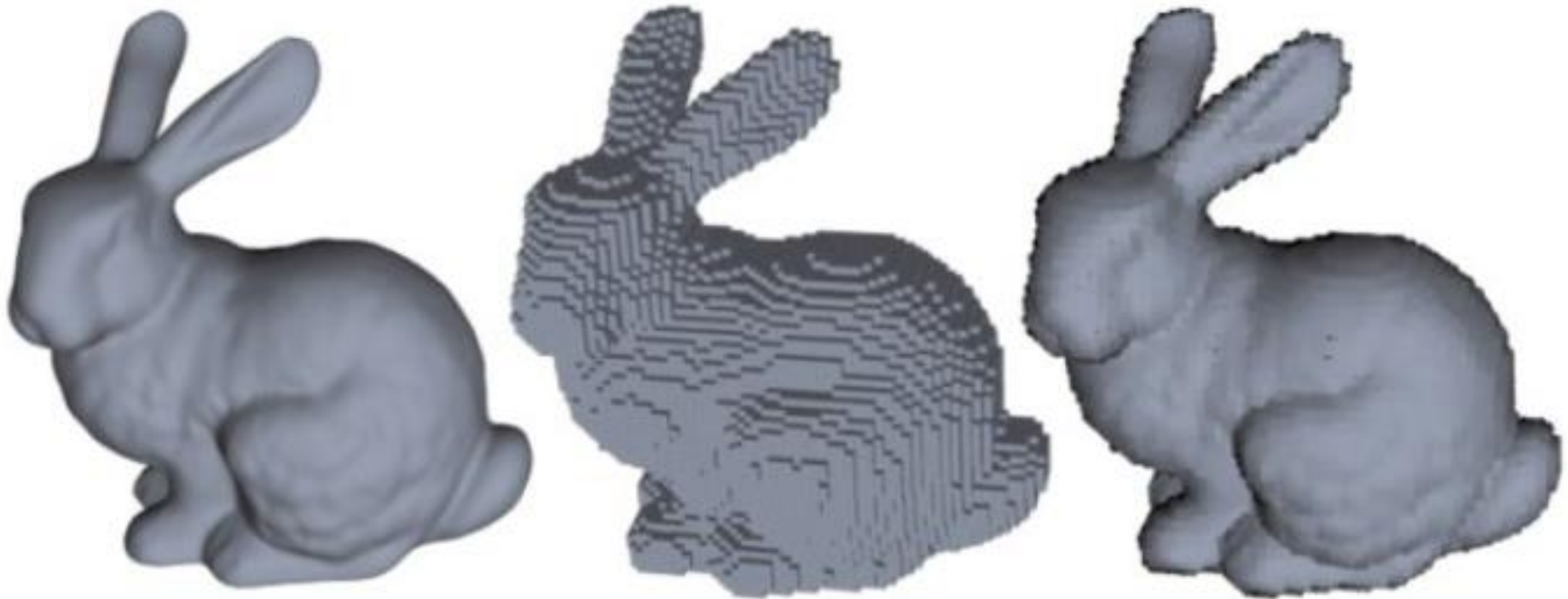
Fig. Left: the original 3D shape. Middle: the voxelized 3D shape. Right: the octree representation with normal sampled at the finest leaf octants.

Reference: O-CNN: Octree-based Convolutional Neural Networks for 3D Shape Analysis , July 2017

[ACM Transactions on Graphics](#) 36(4):1-11 [Peng-Shuai Wang](#), [Yang Liu](#), [Yu-Xiao Guo](#), [Xin Tong](#)

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5.2 Shape vs. Appearance



Chapter 5. Object Recognition from Visual Appearance

5.2 Shape vs. Appearance

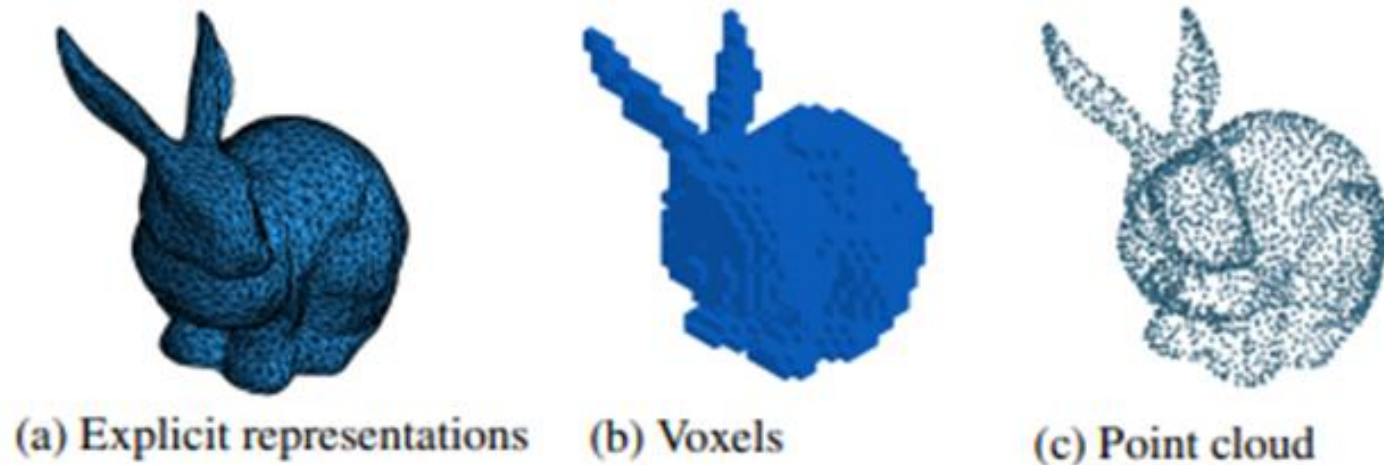


Fig. Common representations of 3D shape.

Reference: <https://arxiv.org/pdf/1901.06802.pdf>

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5.2 Shape vs. Appearance

Basic approaches to represent 3D object.

Representation of the shape of the object. Many approaches have been proposed:

Planes, spheres, complex splines are used
super quadric (see figure).

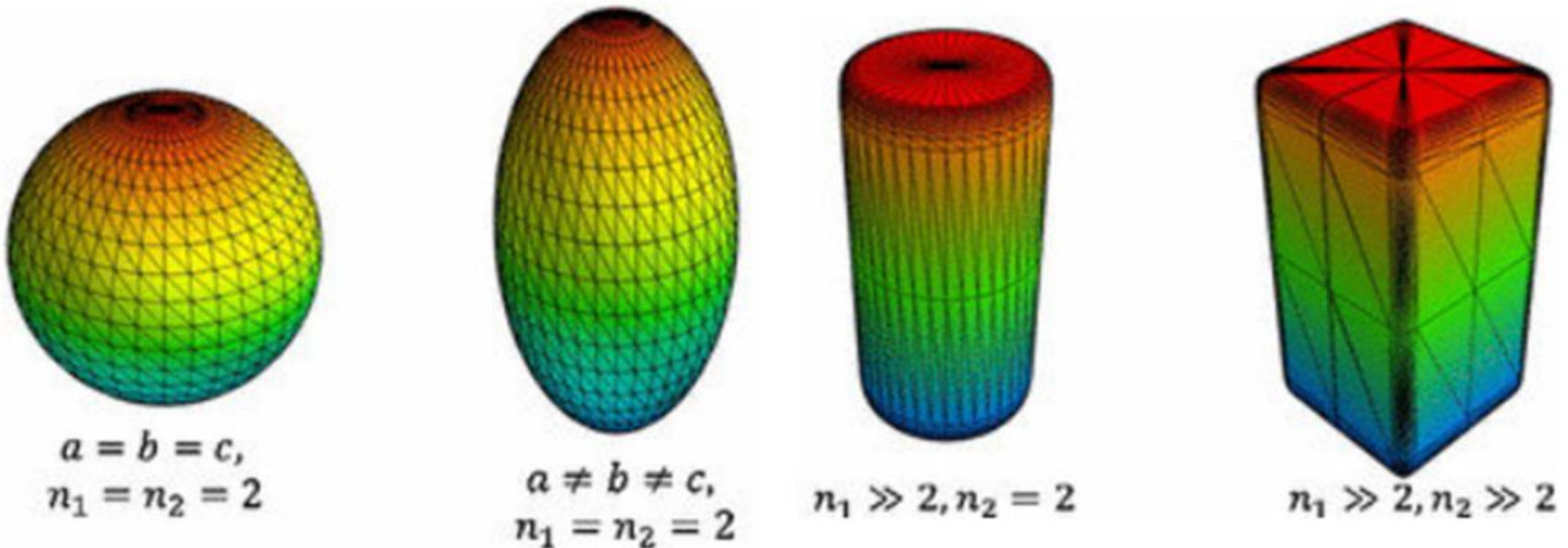
Reference: Efficient implementation of superquadric particles in Discrete Element Method within an open-source framework September 2016 Computational Particle Mechanics 4(1)

DOI: 10.1007/s40571-016-0131-6 Alexander Podlozhnyuk, Stefan Pirker, Christoph Kloss

$$|x|^r + |y|^s + |z|^t = 1, r, s, t \text{ are the parameters to change}$$

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5.2 Shape vs. Appearance



Four examples of superquadric particle shape composed by different shape parameters (a, b, c, n_1, n_2)

$$f(\mathbf{x}) \equiv \left(\left| \frac{x}{a} \right|^{n_2} + \left| \frac{y}{b} \right|^{n_2} \right)^{n_1/n_2} + \left| \frac{z}{c} \right|^{n_1} - 1 = 0,$$
$$\mathbf{x} = (x, y, z)^T,$$

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5.2 Shape vs. Appearance

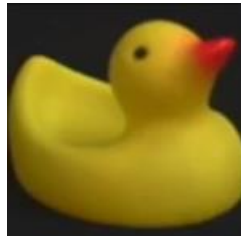
3D Shape matching requires:

- Creating the database of object shapes (offline)
- Recognition (online)
- Computationally expensive for large databases, require hardware (scanners).

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5.2 Shape vs. Appearance

Recognition by visual appearance:
(intrinsic: shape, reflectance,
Extrinsic: pose, illumination)



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5.3 Learning Appearance

When we see an object for the first time, we need to see it from different points of views so as it will be possible to recognize it if we see it again.

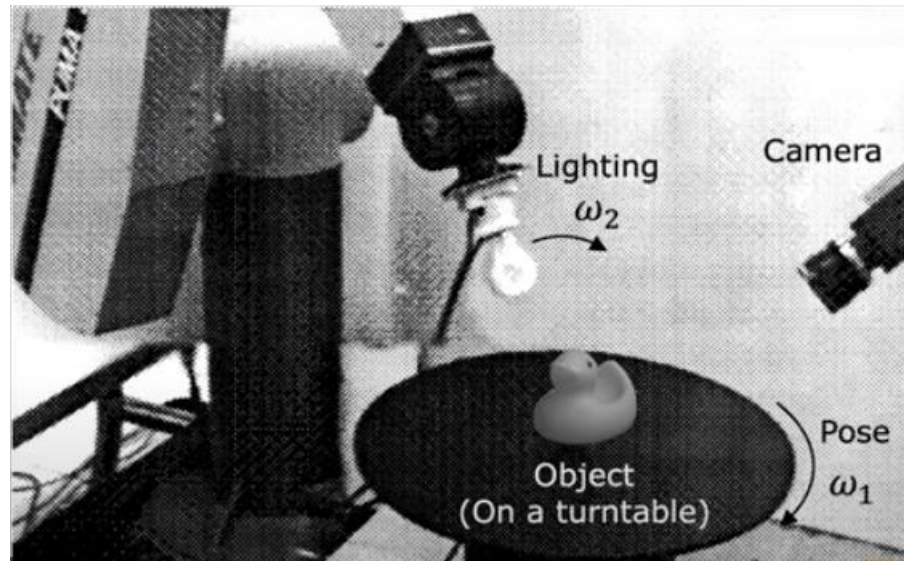


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5.3 Learning Appearance

For a computer vision system, we need to capture images under all poses and lighting directions.

We collect then a set of images (database) , segmented and without occlusion.



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5.3 Learning Appearance

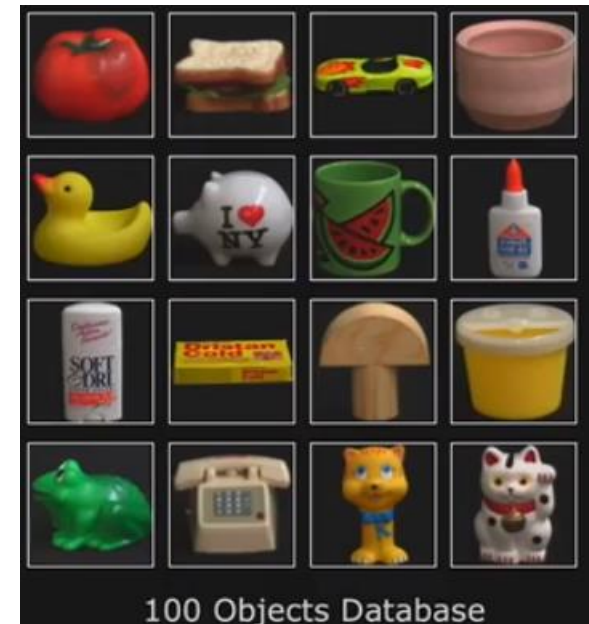
For a computer vision system, we need to capture images under all poses and lighting directions.

We collect then a set of images (database) , segmented and without occlusion.

Example of databases:



M.A. Turk et al, Face recognition using eigenfaces, CVPR 1991



S.K.Nayar et al, Real-Time 100 Object Recognition System ICRA, 1996

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5.3 Learning Appearance

We scale the object in terms of size. We have the object data set.



Nayar , 1996

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5.3 Learning Appearance

We scale the object in terms of size. We have the object data set.



Chapter 5. Object Recognition from Visual Appearance

5.3 Learning Appearance

We scale the object in terms of size. We have the object data set.



Chapter 5. Object Recognition from Visual Appearance

5.3 Learning Appearance

How do we perform recognition?

Naive recognition: Template matching using the similarity measures: SAD (Sum Absolute Distances), or SSD (Sum Squared Differences) or NCC (Normalized Cross Correlation) .



Image query



Object image sets (Template)

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5.3 Learning Appearance

How do we perform recognition?

Naive recognition: Template matching.

This is not a practical approach given the number of images that we are dealing with here.

Template matching is then very expensive and time consuming.



Image query



Object image sets (Template)

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5.3 Learning Appearance

For a given object image set, there is a similarity between two consecutive images? There is a high redundancy between images.



Object image set

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5.3 Learning Appearance

We can exploit the high redundancy between images to reduce the dimensionality of the image set.

We reduce the dimensionality of the image set to achieve a compact appearance representation that makes matching efficient.



Object image set

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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

We want to transform the images into different space where matching one image with another is more efficient.



Image query



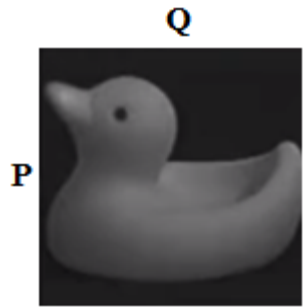
Object image sets

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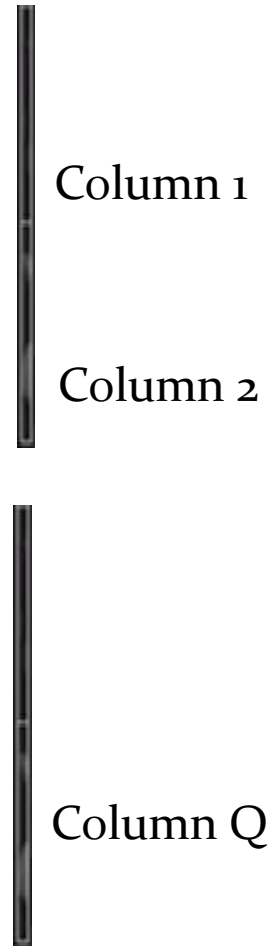
5.4 Dimensionality reduction: PCA

(Principal Component Analysis)

Vector representation of image.



2D image $I(P \times Q = N \text{ pixels})$



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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

We can construct an N-dimensional space. Each one of the dimensions space represents the brightness at corresponding pixel.

The N pixels are represented using N vectors units. The image is simply a point in that space.

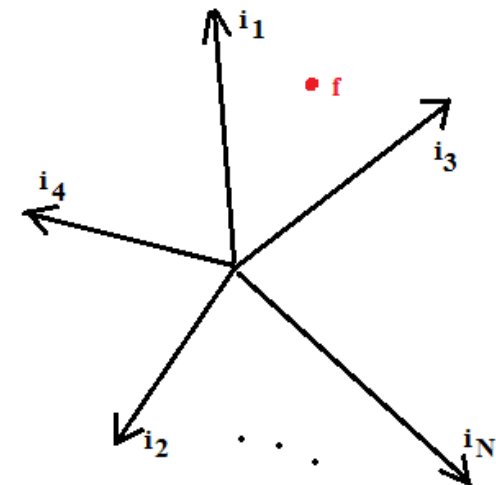
We treat $i_1, i_2, \dots, i_N$ as an orthonormal basis.

Ex: $i_3 = (0, 0, 1, 0, \dots, 0)$

$(123, 0, 0)$

$(120, 20, 0)$

$$SSD = (123 - 120)^2 + (20 - 0)^2$$



N Dimensional space

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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

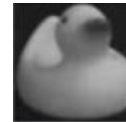
Correlation in Image space:

Image 1



$$SSD = \sum_p \sum_q (I_1(p, q) - I_2(p, q))^2$$

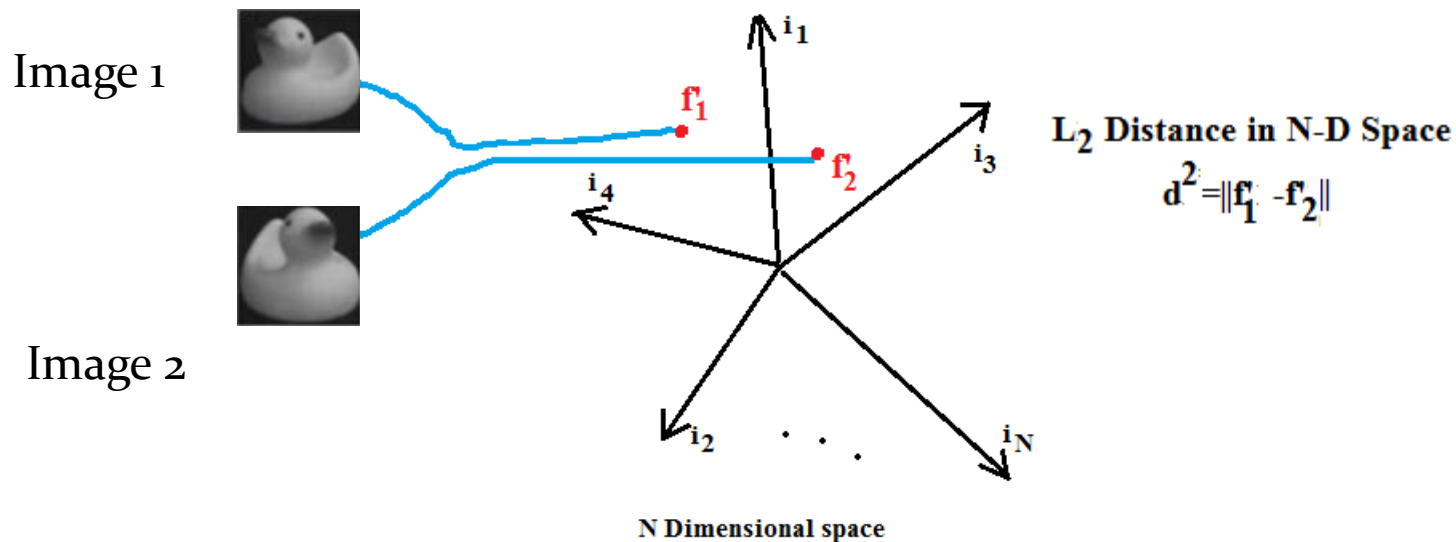
Image 2



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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Correlation in Image space:



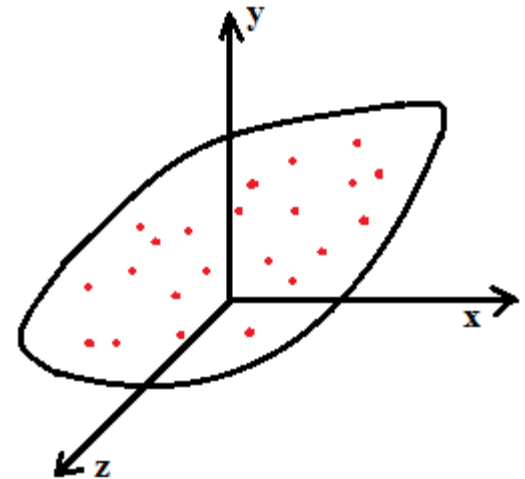
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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Dimensionality reduction:

Consider a distribution of 3D points that lie on a 2D plane.

It is redundant to represent each point with 3 coordinates.

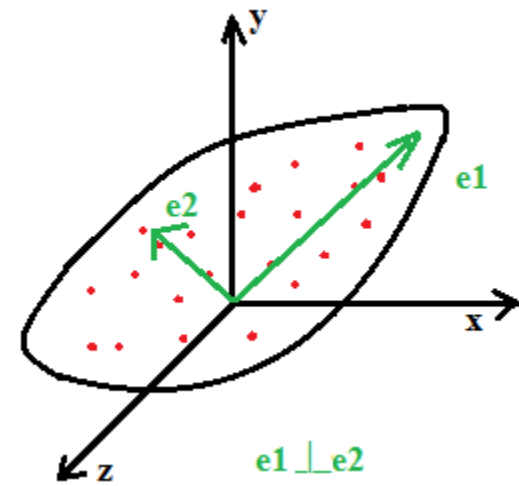


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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Dimensionality reduction:

If we use a new coordinates system (e_1 , e_2) that lies on the plane, each point can be represented with just 2 coordinates.

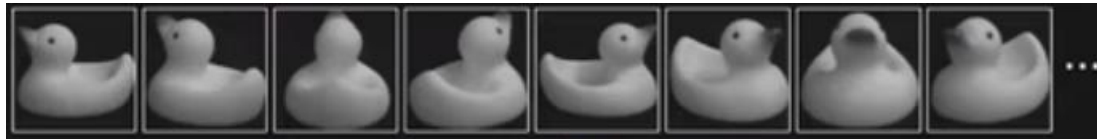


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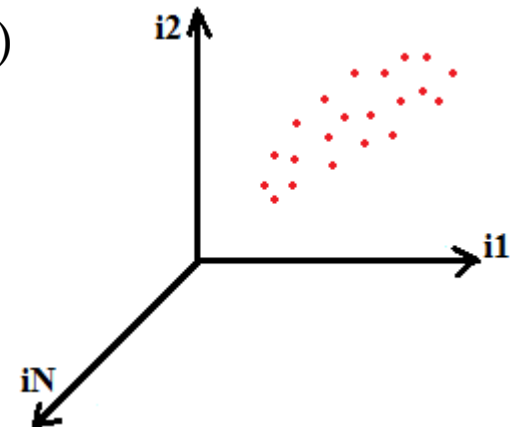
5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Appearance distribution:

For M images, we have in the N dimensional space, the distribution (see below). This is called Appearance Distribution.



Object Image set (M images)

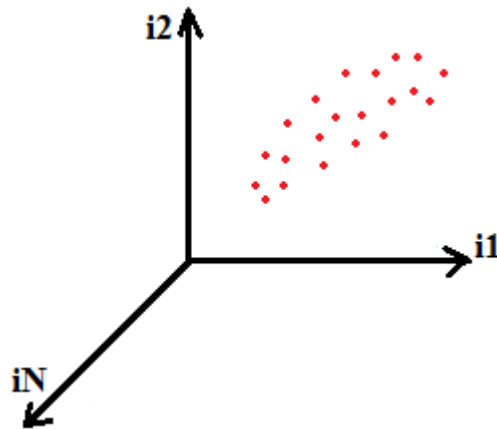


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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Appearance distribution:

Because there is correlation between images, the distribution of feature points is highly structured and usually resides in small number of dimensions.



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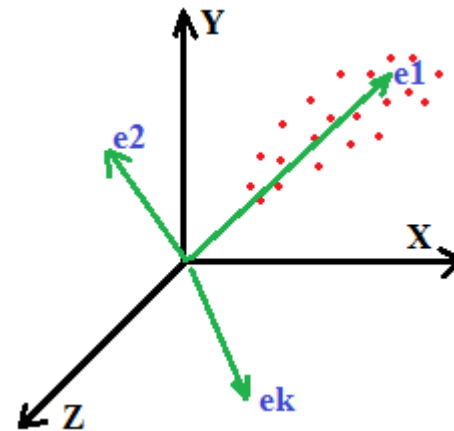
5.4 Dimensionality reduction: PCA

(Principal Component Analysis)

Appearance distribution:

It is possible to express the feature point distribution in a lower dimension space ($k < N$).

How to calculate this basis ($e_1, \dots, e_k$)?



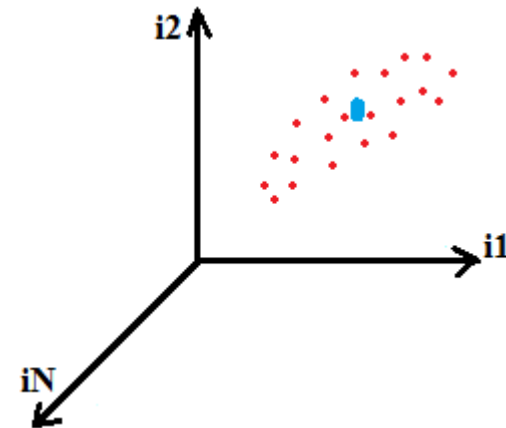
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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Subtracting the mean:

Given M images ($f'_1, f'_2, \dots, f'_M$) of an object, the Mean Image is:

$$c = \frac{1}{M} \sum_{m=1}^M f'_m$$



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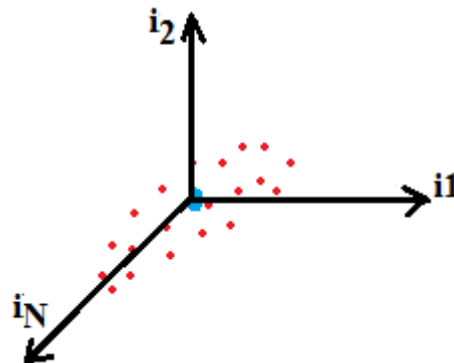
5.4 Dimensionality reduction: PCA

(Principal Component Analysis)

Subtracting the mean:

Subtract the mean from the object image set so as to move the origin of the new basis to the centroid of the distribution.

$$f_m = f'_m - c$$



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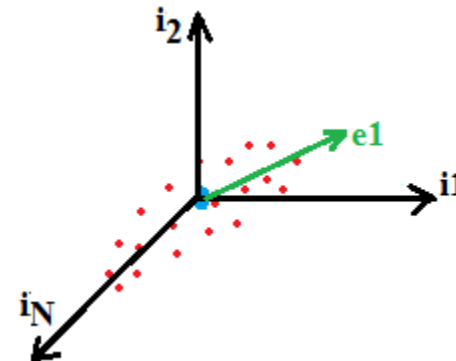
5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Principal components:

First Principal component (e_1):

The direction of maximum variance in the image set.

$$V = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$



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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

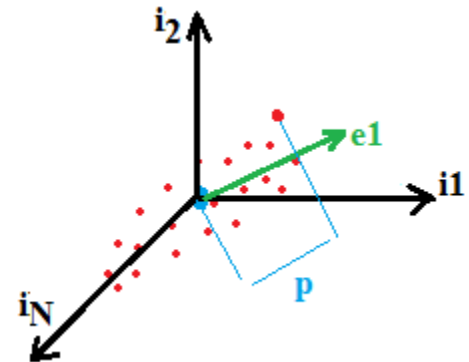
Principal components:

Image representation:

Project the image onto the principal component e_1 .

$$p = e_1 \cdot f \text{ (dot product)}$$

Image is represented by a single number p .



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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

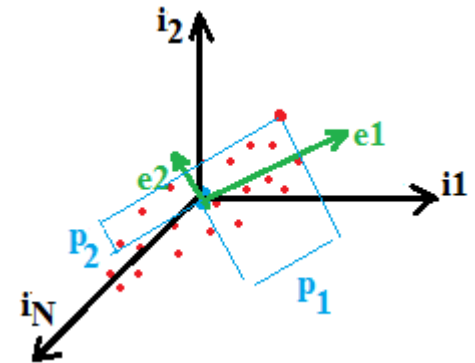
Principal components:

The second principal component e_2 :
Is the direction of the second maximum variance in
the image set such that:

e_1 perpendicular e_2

Image is represented by a two numbers p_1, p_2 .

$$p = \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = [e_1 \ e_2]^T f$$



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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Principal components:

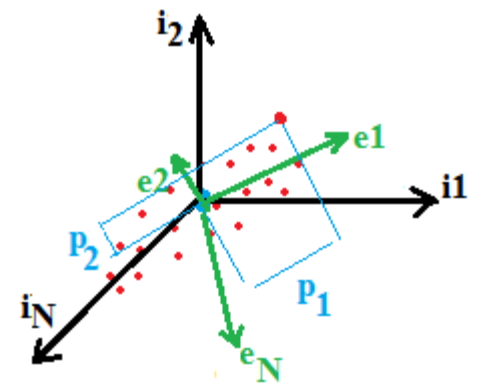
The k^{th} principal component e_k :

Is the direction of the second maximum variance in the image set such that:

e_1 perpendicular e_2 perpendicular e_k

Image is represented by a k numbers $p_1, p_2, \dots, p_k$ (in $k < N$ dimensions): Forward projection.

$$p = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_k \end{bmatrix} = [e_1 \ e_2 \ \dots \ e_k]^T f$$



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5.4 Dimensionality reduction: PCA (Principal Component Analysis)

Principal components:

The k^{th} principal component e_k :

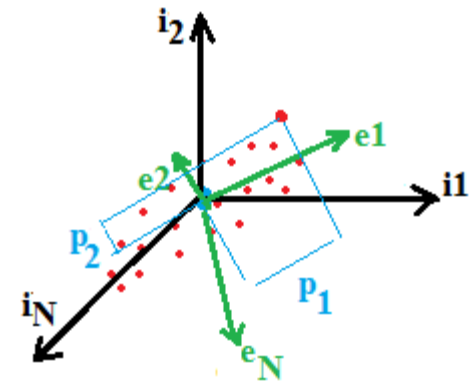
Is the direction of the second maximum variance in the image set such that:

e_1 perpendicular e_2 perpendicular e_k

Back projection.

$$f \approx \sum_1^K p_k e_k$$

$(e_1, e_2, \dots, e_k)$ is referred to as **Linear Subspace**.



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5.5 Finding Principal Components

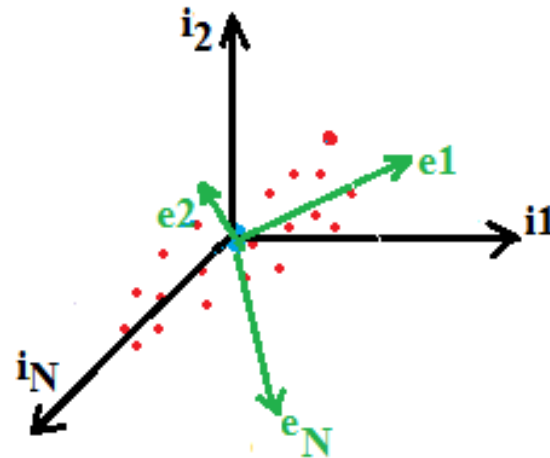
Given the Mean-Subtracted Image Set $(f_1, f_2, \dots, f_M)$, f_M is $N \times 1$ vector

Find the orthogonal basis $(e_1, e_2, \dots, e_K)$, e_K is $N \times 1$ vector.

Such that:

$$f_m \approx \sum_1^K p_k^{(m)} e_k$$

$$p_k^{(m)} = e_k^T f_m$$



$$e_i^T e_j = 1 \text{ if } i = j \text{ and } 0 \text{ if } i \neq j$$

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5.5 Finding Principal Components

Projection of the image f along the first principal component $\mathbf{e}$ is $f \cdot \mathbf{e}$

Find $\mathbf{e}$ that maximizes Variance of the $f \cdot \mathbf{e}$

$$\text{Var} [f \cdot \mathbf{e}] = \mathbf{E}[\{(f \cdot \mathbf{e} - \mathbf{E}[f \cdot \mathbf{e}])\}^2]$$

Where $\mathbf{E}[X]$ is the expected value of X .

$$\mathbf{E}[X] = \sum X \cdot P(X)$$

$$\text{Var} [f \cdot \mathbf{e}] = \mathbf{E}[\{\mathbf{e} \cdot (f - \mathbf{E}[f])\}^2], \mathbf{E}[f]=0$$

$$\text{Var} [f \cdot \mathbf{e}] = \mathbf{E}[\{(e \cdot f)\}^2] = \mathbf{E}[\mathbf{e}^T \mathbf{f} \mathbf{f}^T \mathbf{e}]$$

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5.5 Finding Principal Components

Find $\mathbf{e}$ that maximizes $[\mathbf{e}^T \mathbf{f} \mathbf{f}^T \mathbf{e}]$ such that $\mathbf{e}^T \mathbf{e} = 1$

Maximize Objective function $L(e, \lambda)$ such that:

$$L(e, \lambda) = e^T R e - \lambda(e^T e - 1)$$

Taking derivatives of $L(e, \lambda)$ w.r.t. e and equating to zero:

$$R e - \lambda e = 0, R e = \lambda e, \text{EigenValue problem.}$$

The first principal Component is eigenvector corresponding to maximum eigenvalue.

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5.5 Finding Principal Components

The algorithm:

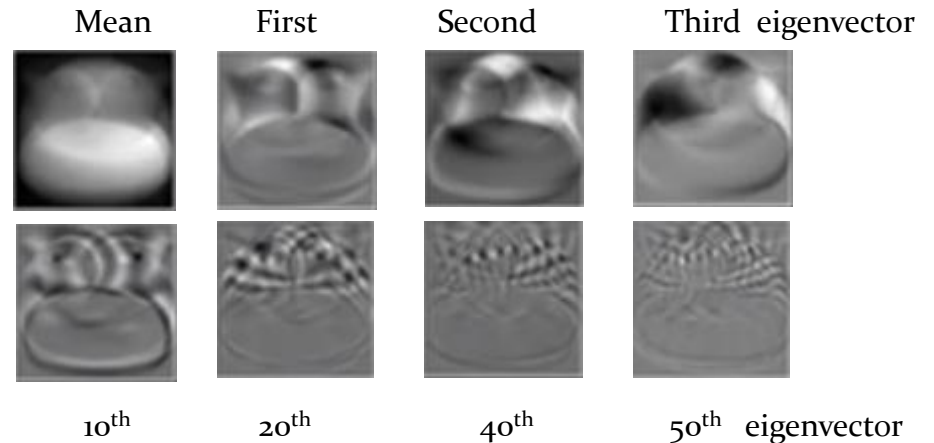
- Data Matrix: $F = [f_1, f_2, \dots, f_M]$
- Covariance Matrix $R = FF^T$
- $X_i = X_j = f_e$
 $\text{cov}[X_i, X_j] = E[(X_i - E[X_i])(X_j - E[X_j])]$
- Solve EigenValue problem: $Re = \lambda e$
- Eigenvalues: $\lambda_1, \lambda_2, \dots, \lambda_K$
- Eigenvectors: $e_1, e_2, \dots, e_K$

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5.6 Parametric Appearance Representation

Dimensionality Reduction:

From an object image set, we get K Eigenvectors (see example below:

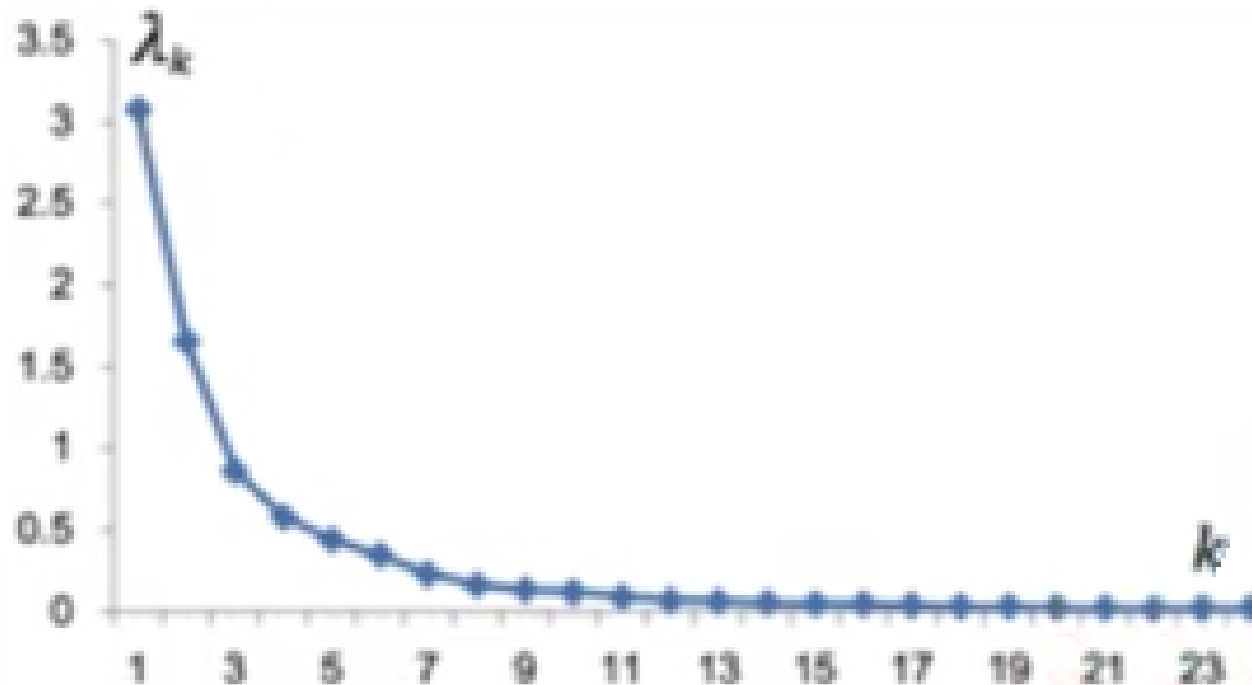


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5.6 Parametric Appearance Representation

Dimensionality Reduction:

From an object image set, we get K Eigenvectors (see example below:



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5.6 Parametric Appearance Representation

How many principal components (K) are sufficient?

If we want to capture 95% of variations in the data set, then, find smallest K such that:

$$\frac{\text{Sum of } K \text{ largest Eigenvalues}}{\text{Sum of All EigenValues}} \geq 0.95$$

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5.7 Appearance Matching

Algorithm 1: Dataset representation

Given M learning images $\{I_1^{(q)}, I_2^{(q)}, \dots, I_M^{(q)}\}$ for each object q (=1..Q) of Q training objects

1- Normalize all images to remove brightness variations:

$$I_m'^{(q)} = I_m^{(q)} / \|I_m^{(q)}\|$$

2- Convert image $I_m'^{(q)}$ to a vector $f_m'^{(q)}$

3- Compute the mean vector $c^{(q)}$ of each object.

4- Subtract the mean feature vector $c^{(q)}$ for object q

$$f_m^{(q)} = f_m'^{(q)} - c^{(q)}$$

5- Construct the data matrix and covariance matrix:

$$F^{(q)} = \begin{bmatrix} f_1^{(q)} & f_2^{(q)} & \dots & f_M^{(q)} \end{bmatrix}$$
$$R^{(q)} = F^{(q)} F^{(q)T}$$

6- Compute the K eigenvectors $e_1^{(q)}, e_2^{(q)}, \dots, e_K^{(q)}$ of $R^{(q)}$

7- Project feature vector to eigenvectors for **object q**:

$$p_m^{(q)} = [e_1^{(q)}, e_2^{(q)}, \dots, e_K^{(q)}]^T \times f_m^{(q)}$$

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5.7 Appearance Matching

Algorithm 2: Object recognition

Given input image (I) for object recognition

1- Normalize the image to remove brightness variations:

$$I' = I / \|I\|$$

2- Convert image I' to a vector f'

For each object q in the database, perform steps 3-6:

3- Compute the mean vector $c^{(q)}$ of each object.

4- Subtract the mean feature vector $c^{(q)}$ for object q

$$f_q = f' - c^{(q)}$$

5- Project feature vector to eigenspace for object q :

$$p^q = [e_1^{(q)}, e_2^{(q)}, \dots, e_K^{(q)}]^T \times f^{(q)}$$

6- In the eigenspace of object q find the closest point to projected point, compute the distance $d^{(q)}$.

7- Find the object for which $d^{(q)}$ is minimum.

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5.8 Example of Applications

Face recognition using eigenfaces

Reference: Kirvansh Kallaram, [ICT University of Mauritius](https://www.youtube.com/watch?v=61NuFlK5VdU)

<https://www.youtube.com/watch?v=61NuFlK5VdU>

1- Obtain face images $I_1, I_2, \dots, I_m$ as training dataset faces

Example: $I_1 = \begin{bmatrix} 1 & 1 \\ -2 & -3 \end{bmatrix}, I_2 = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}, I_3 = \begin{bmatrix} 2 & -2 \\ 1 & 3 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$

2- Represent each image as a vector

$$I'_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \\ -3 \end{bmatrix}, I'_2 = \begin{bmatrix} 1 \\ -1 \\ 3 \\ 2 \end{bmatrix}, I'_3 = \begin{bmatrix} 2 \\ -2 \\ 1 \\ 3 \end{bmatrix}, I'_4 = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 1 \end{bmatrix}$$

3- Mean vector c

$$c = \frac{1}{4} \begin{bmatrix} 1 + 1 + 2 + 1 \\ 1 - 1 - 2 + 2 \\ -2 + 3 + 1 + 2 \\ -3 + 2 + 3 + 1 \end{bmatrix} = \begin{bmatrix} 5/4 \\ 1 \\ 0 \\ 3/4 \end{bmatrix},$$

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5.8 Example of Applications

Face recognition using eigenfaces

1- Difference between each image and the average

$$I'_1 - c = \begin{bmatrix} -1/4 \\ 0 \\ -2 \\ -15/4 \end{bmatrix}, I'_2 - c = \begin{bmatrix} -1/4 \\ 0 \\ 3 \\ 5/4 \end{bmatrix}, I'_3 - c = \begin{bmatrix} 3/4 \\ -3 \\ 1 \\ 9/4 \end{bmatrix}, I'_4 - c = \begin{bmatrix} -1/4 \\ 1 \\ 2 \\ 1/4 \end{bmatrix}$$

3- Covariance matrix = $\frac{1}{m} F F^T$

$$F = \begin{bmatrix} -\frac{1}{4} & -\frac{1}{4} & \frac{3}{4} & -\frac{1}{4} \\ 0 & 0 & -3 & 1 \\ -2 & 3 & 1 & 2 \\ -\frac{15}{4} & \frac{5}{4} & \frac{9}{4} & \frac{1}{4} \end{bmatrix}, n^2 \times m \text{ (where } n \times n \text{ is the image dimension)}$$

4- Compute the eigenvalues of $F^T F$ in order to have the dimension $m \times m$

$$\lambda_1 = 0, \lambda_2 = 0,374, \lambda_3 = 2,208, \lambda_4 = 8,793$$

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5.8 Example of Applications

Face recognition using eigenfaces

5- Compute the eigen vectors

$$V_1 = \begin{bmatrix} -3 \\ -1 \\ 0 \\ 1 \end{bmatrix}, V_2 = \begin{bmatrix} 0.63 \\ -0.9 \\ 0.8 \\ 1 \end{bmatrix}, V_3 = \begin{bmatrix} 4.6 \\ -13 \\ -17.76 \\ 1 \end{bmatrix}, V_4 = \begin{bmatrix} 0.09 \\ 0.72 \\ -0.45 \\ 1 \end{bmatrix}$$

The eigenvectors:

v_1, v_2, v_3, v_4 are normalized such that $||v_i||=1$

$$||v_1||=\sqrt{12}$$

$$||v_2||=\sqrt{3}$$

$$||v_3||=\sqrt{509}$$

$$||v_4||=\sqrt{1.73}$$

We compute $U_i = V_i / ||v_i||$, U_i are the eigenfaces.

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5.8 Example of Applications

Face recognition using eigenfaces

The eigenvectors:

u_1, u_2, u_3, u_4 are normalized such that $||v_i||=1$

Any image q_i of the dataset is represented by the matrix Q_i :

$$Q_i = c \text{ (mean faces)} + \sum_{m=1}^K u_m w_m$$

w_m are weights (contribution of each eigenvector), is the product of image (vector) with each of the eigenvectors.

w_m is computed as the product of the image with the eigen vector.

$$w_k = U_k^T I'_k$$

If $m=k$, we select the k most important eigen vectors.

Chapter 5. Object Recognition from Visual Appearance

5.8 Example of Applications

Face recognition using eigenfaces

To perform recognition of a test image I_t .

We compute the weight of the test image $I_{t'} = I_t - c$

$$w'_k = U_k^T I_{t'}, \text{ for } k=1 \text{ to } m \text{ (} m=4 \text{)}$$

We compute after the euclidean distance between

(w'_1, w'_2, w'_3, w'_4) and the weights of each class: (w_1, w_2, w_3, w_4)

And choose the class with a minimum distance.